

1. App Proposal

App name

SkinTect

What the app is for, in one sentence

A tool that lets a user photograph a skin lesion and get an informal read on whether it looks benign or malignant, with a confidence score and visual explanation as a supplement to, not a replacement for, seeing a dermatologist.

Who is it for

Primary: The app is primarily for me and for anyone looking at my project portfolio. It showcases my experience in ML, JS, React, and such.

Secondary: Anyone who wants to do a quick, informal check before deciding whether it's worth booking a visit to a dermatologist.

Sections or routes this app needs

#	Section / Route	What is it for
1	Login / Sign in	User registration or login
2	Home	Upload/Take photo, view prediction + heatmap + confidence score
3	Profile	User can view, edit, and delete account
4	History	View log of previous scans and results

State: what data does the app hold?

Data	Shape (rough)	Who owns it (which component)	Changes when...
uploadedImage	File/image URL string	Home	User uploads or captures a new photo
prediction	{ label: "benign" "malignant", confidence: num }	Home	Model returns a new result after image upload
heatmap	image URL string	Home	Prediction is generated
scanStatus	{ "idle" "loading" "finished" "error" }	Home	Upload starts, model responds, or the generation fails

scanHistory	[{ id, imageUrl, label, confidence, date }]	History	Completion of a new scan is saved
currentUser	{ id, name, email }	App	User logs in or out, or profile is edited

What each screen contains

Home

- Block 1: Upload/Take photo button (file input or camera capture)
- Block 2: Preview of the uploaded image
- Block 3: Submit button
- Block 4: Results panel (e.g., prediction label, confidence score, heatmap)
- Block 5: Loading state while the model runs

Login/Sign In

- Block 1: Email and password fields
- Block 2: Submit button
- Block 3: Error message
- Block 4: Switch between login and sign in modes

History

- Block 1: List of past scans, showing a thumbnail, label, confidence score, and date
- Block 2: Clickable icon to view detailed scan (e.g., full image, heatmap)
- Block 3: Empty State (no scans done yet)
- Block 4: Filter (e.g., by date or by label)

Profile

- Block 1: Display of current user info (name, email)
- Block 2: Editing form for updating name, email, password
- Block 3: Save/cancel options with confirmation feedback
- Block 4: Delete account button with confirmation feedback

Content you need to gather

- Trained model
- Logo
- Sample lesion images for testing during development
- Medical disclaimer text block and explicit warnings (e.g., **“NOT A DIAGNOSIS”**)
- Database and configurations

One risk

- Wiring the React frontend to the model inference. Getting the heatmap generated accurately on top of the uploaded image, since it depends on how the model outputs it.